



《数值分析》17

主要内容： 函数逼近与希尔伯特矩阵

切比雪夫多项式

勒让德多项式

正交多项式的应用

函数逼近与希尔伯特矩阵

问题. 求二次多项式 $P(x) = a_0 + a_1x + a_2x^2$ 使

$$\int_0^1 [P(x) - \sin(\pi x)]^2 dx = \min$$

连续函数的最佳平方逼近

已知 $f(x) \in C[0, 1]$, 求多项式

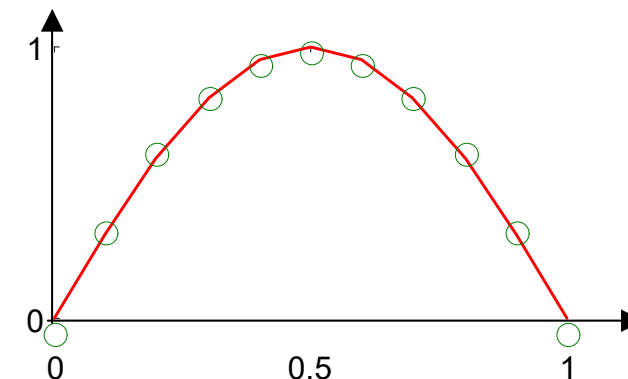
$$P(x) = a_0 + a_1x + a_2x^2 + \dots + a_n x^n$$

使得

$$L = \int_0^1 [P(x) - f(x)]^2 dx = \min$$

$$\text{令 } L(a_0, a_1, \dots, a_n) = \int_0^1 \left[\sum_{j=0}^n a_j x^j - f(x) \right]^2 dx$$

$$L = \int_0^1 \left[\sum_{j=0}^n a_j x^j \right]^2 dx - 2 \sum_{j=0}^n a_j \int_0^1 x^j f(x) dx + \int_0^1 [f(x)]^2 dx$$



$$\frac{\partial L}{\partial a_k} = 2 \sum_{j=0}^n a_j \int_0^1 x^{j+k} dx - 2 \int_0^1 x^k f(x) dx$$

$$\text{令 } \frac{\partial L}{\partial a_k} = 0 \quad \text{记 } b_k = \int_0^1 x^k f(x) dx$$

$$\begin{bmatrix} 1 & 1/2 & \cdots & 1/(n+1) \\ 1/2 & 1/3 & \cdots & 1/(n+2) \\ \cdots & \cdots & \cdots & \cdots \\ 1/(n+1) & \cdots & \cdots & 1/(2n+1) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_n \end{bmatrix}$$

系数矩阵被称为**Hilbert**矩阵

定义6.3 设 $f(x), g(x) \in C[a, b]$, $\rho(x)$ 是区间 $[a, b]$ 上的权函数, 若等式

$$(f, g) = \int_a^b \rho(x) f(x) g(x) dx = 0$$

成立, 则称 $f(x), g(x)$ 在 $[a, b]$ 上带权 $\rho(x)$ 正交.

当 $\rho(x)=1$ 时, 简称正交。

例1 验证 $\varphi_0(x)=1$, $\varphi_1(x)=x$ 在 $[-1, 1]$ 上正交,
并求二次多项式 $\varphi_2(x)$ 使之与 $\varphi_0(x), \varphi_1(x)$ 正交

解:
$$\int_{-1}^1 \varphi_0(x) \varphi_1(x) dx = \int_{-1}^1 1 \cdot x dx = 0$$

$$\text{设 } \varphi_2(x) = x^2 + a_{21}x + a_{22}$$

$$\int_{-1}^1 1 \cdot \varphi_2(x) dx = 0 \qquad \int_{-1}^1 x \varphi_2(x) dx = 0$$

$$\int_{-1}^1 (x^2 + a_{21}x + a_{22}) dx = 0 \quad \int_{-1}^1 x(x^2 + a_{21}x + a_{22}) dx = 0$$

$$2/3 + 2a_{22} = 0$$

$$2a_{21}/3 = 0$$

$$a_{22} = -1/3$$

$$a_{21} = 0$$

$$\text{所以, } \varphi_2(x) = x^2 - \frac{1}{3}$$

切比雪夫多项式:

$$T_0(x)=1, T_1(x)=\cos\theta = x, T_2(x)=\cos 2\theta \cdots \cdots$$

$$T_n(x)=\cos(n\theta), \cdots \cdots \cdots$$

1. 递推公式

由 $\cos(n+1)\theta = 2 \cos\theta \cos(n\theta) - \cos(n-1)\theta$ 得

$$T_{n+1}(x) = 2x T_n(x) - T_{n-1}(x) \quad (n \geq 1)$$

所以, $T_0(x)=1, T_1(x)=x, T_2(x)=2x^2 - 1, \cdots \cdots \cdots$

2.切比雪夫多项式的正交性

$$\int_0^{\pi} \cos(m\theta) \cos(n\theta) d\theta = 0$$

$$\begin{aligned}(T_m, T_n) &= \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} T_m(x) T_n(x) dx \\ &= \int_0^{\pi} \cos m\theta \cos n\theta d\theta = 0\end{aligned}$$

所以, 切比雪夫多项式在 $[-1, 1]$ 上带权

$$\rho(x) = \frac{1}{\sqrt{1-x^2}} \quad \text{正交}$$

勒让德(Legendre)多项式

1. 表达式 $P_0(x) = 1, P_1(x) = x$

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} [(x^2 - 1)^n] \quad (n \geq 1)$$

2. 正交性

$$\int_{-1}^1 P_m(x) P_n(x) dx = \begin{cases} 0, & m \neq n \\ \frac{2}{2n+1}, & m = n \end{cases}$$

3.递推式

$$\begin{cases} p_0 = 1, & p_1 = x, \\ p_{n+1} = \frac{2n+1}{n+1}xp_n - \frac{n}{n+1}p_{n-1} \end{cases}$$
$$p_2(x) = \frac{1}{2}(3x^2 - 1) \quad p_3(x) = \frac{1}{2}(5x^3 - 3x)$$

4.零点分布

$P_n(x)$ 的 n 个零点,落入区间 $[-1, 1]$ 中

$$P_2(x) \text{ 的两个零点: } x_1 = -\frac{1}{\sqrt{3}} \quad x_2 = \frac{1}{\sqrt{3}}$$

$$P_3(x) \text{ 的三个零点: } x_1 = -\sqrt{\frac{3}{5}} \quad x_2 = 0 \quad x_3 = \sqrt{\frac{3}{5}}$$

用正交多项式作最佳平方逼近

设 $P_0(x), P_1(x), \cdots, P_n(x)$ 为区间 $[a, b]$ 上的正交多项式, 即

$$(P_k, P_j) = \int_a^b P_k(x) P_j(x) dx = 0 \\ (k \neq j, k, j = 0, 1, \cdots, n)$$

求 $P(x) = a_0 P_0(x) + a_1 P_1(x) + \cdots + a_n P_n(x)$

使 $L = \int_a^b [P(x) - f(x)]^2 dx = \min$

$$L(a_0, a_1, \cdots, a_n) = \int_a^b \left[\sum_{j=0}^n a_j P_j(x) - f(x) \right]^2 dx$$

$$\frac{\partial L}{\partial a_k} = 2 \int_a^b P_k(x) \left[\sum_{j=0}^n a_j P_j(x) - f(x) \right] dx$$

由于 $(P_k, P_j) = \int_a^b P_k(x) P_j(x) dx = 0, (k \neq j)$

令 $\frac{\partial L}{\partial a_k} = 0$ 记 $(P_k, f) = \int_a^b P_k(x) f(x) dx$

则有 $(P_k, P_k) a_k = (P_k, f)$

$$a_k = \frac{(P_k, f)}{(P_k, P_k)} \quad (k = 0, 1, 2, \dots, n)$$

$f(x)$ 的平方逼近 $P(x) = \sum_{k=0}^n \frac{(P_k, f)}{(P_k, P_k)} P_k(x)$

例 求二次多项式 $P(x) = a_0 + a_1x + a_2x^2$ 使

$$\int_0^1 [P(x) - \sin(\pi x)]^2 dx = \min$$

构造区间 $[0, 1]$ 上的正交多项式

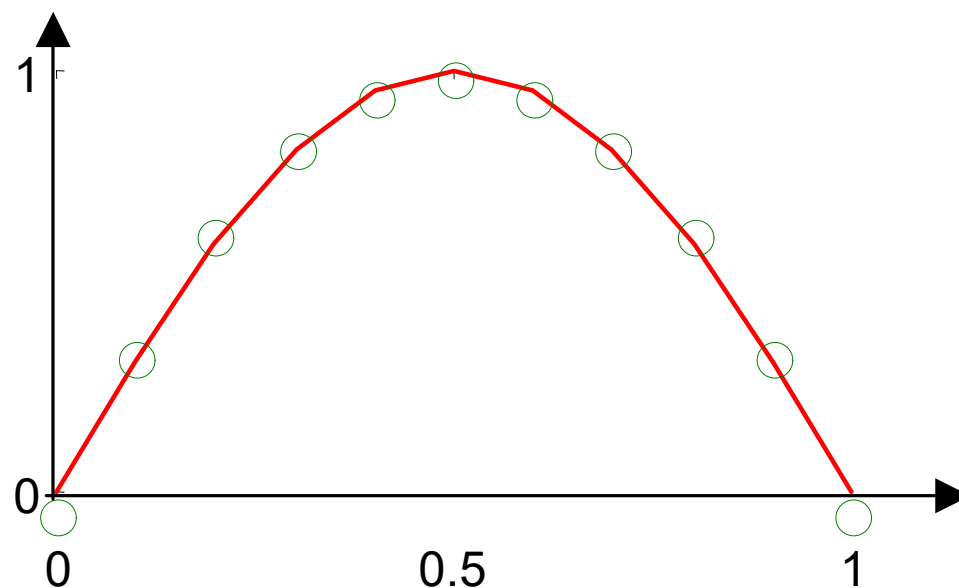
$$P_0(x) = 1, \quad P_1(x) = x - 1/2, \quad P_2(x) = x^2 - x + 1/6$$

$$\sin(\pi x) \approx \frac{(P_0, \sin(\pi x))}{(P_0, P_0)} + \frac{(P_1, \sin(\pi x))}{(P_1, P_1)} P_1(x) + \frac{(P_2, \sin(\pi x))}{(P_2, P_2)} P_2(x)$$

$$\frac{(P_0, \sin(\pi x))}{(P_0, P_0)} = \frac{2/\pi}{1} \quad \frac{(P_1, \sin(\pi x))}{(P_1, P_1)} = \frac{0}{1/12}$$

$$\frac{(P_2, \sin(\pi x))}{(P_2, P_2)} = \frac{(\pi^2 - 12)/3\pi^3}{1/180}$$

最佳平方逼近: $\sin(\pi x) \approx \frac{2}{\pi} - 4.1225(x^2 - x + \frac{1}{6})$



○ $P(x) = \frac{2}{\pi} - 4.1225(x^2 - x + \frac{1}{6})$

— $f(x) = \sin(\pi x)$

MATLAB符号命令求解

```
syms x
```

```
P1=inline('x-.5');
```

```
P2=inline('x^2-x+1/6');
```

```
c0=int(sin(pi*x),0,1);
```

```
c1=int(P1(x)*sin(pi*x),0,1)/int(P1(x)*P1(x),0,1);
```

```
c2=int(P2(x)*sin(pi*x),0,1)/int(P2(x)*P2(x),0,1)
```

```
numeric([c0,c1,c2])
```

```
ans =    0.6366    0 -4.1225
```

```
P=inline('0.6366-4.1225*(x.^2-x+1/6)')
```

```
t=0:.1:1;y=sin(pi*t);pp=P(t);plot(t,y,t,pp,'o')
```

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